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SECR3553 ARTIFICIAL INTELLIGENCE
(KEPINTARAN BUATAN)
SECTION 03

ASSIGNMENT 1:
PARKING DETECTION APPLICATION

THEME:
SMART CITY

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Knowledge Representation

Knowledge Representation (KR) in this Smart Parking System defines how knowledge is logically modeled to enable intelligent reasoning and automated decision-making. The system uses **IF–THEN rules (Production Rules)** and **First Order Logic (FOL)** to detect parking availability, guide drivers, and maintain real-time data accuracy.

These rules are processed by the AI module using inputs from cameras and sensors, ensuring reliable and efficient parking management in urban areas.

1. Detect Occupied Parking Space

IF vehicle_detected = TRUE THEN space_status = Occupied

Explanation:

When a vehicle is detected in a parking slot through image recognition, the AI system classifies the parking space as occupied. This prevents the system from marking the spot as available to other drivers. The rule ensures data accuracy and helps maintain the system's credibility by providing up-to-date parking information. This KR supports the goal of real-time parking detection and improves the efficiency of managing parking availability in congested areas.

FOL Representation:

$\forall x (\text{ParkingSpace}(x) \wedge \text{VehicleDetected}(x) \rightarrow \text{SpaceStatus}(x, \text{Occupied}))$

For every parking space, if a vehicle is detected, then the space status is "Occupied."

2. Detect Available Parking Space

IF vehicle_detected = FALSE THEN space_status = Available

AND

IF vehicle_detected = FALSE for a continuous period of T seconds THEN confirm space_status = Available

Explanation:

When no vehicle is detected in a parking slot, the system temporarily marks the slot as "Available." However, to prevent false detections due to environmental interference such as poor lighting, shadows, or transient obstructions, the system verifies that the slot remains empty for a defined period T.

This temporal validation rule improves data reliability by filtering short-term errors before updating the system. The AI ensures the information shown to drivers reflects the actual availability of the slot, thus enhancing user trust and reducing false updates.

FOL Representation:

$$\forall x (\text{ParkingSpace}(x) \wedge \neg \text{VehicleDetected}(x) \rightarrow \text{SpaceStatus}(x, \text{Available}))$$

$$\forall x \forall t (\text{ParkingSpace}(x) \wedge \text{NoVehicleDetectedFor}(x, t, T) \rightarrow \text{ConfirmedStatus}(x, \text{Available}))$$

For every parking space x and time t , if no vehicle is detected for a continuous duration T , confirm the space status as Available.

3. Provide Nearest Available Parking Spot

IF driver_request = TRUE THEN display_nearest(space_status = Available)

Explanation:

When a driver requests to find a parking space, the AI system processes real-time data from the database to identify which spaces are available. It then determines the nearest available spot based on the driver's current location and displays it on the interface. This feature helps drivers save time, reduce frustration, and minimize unnecessary traffic movement within parking areas.

FOL Representation:

$$\forall x (\text{DriverRequest}(x) \rightarrow \text{Display}(\text{Nearest}(\text{ParkingSpace}(\text{Available}))))$$

For every driver request, display the nearest available parking space.

4. Update Parking Database

IF space_status changes THEN update database

Explanation:

When the status of a parking space changes (for example, from Available to Occupied or vice versa), the system must immediately update the central database to ensure all users see accurate, real-time parking information. This knowledge rule ensures the synchronization between the detection system and the mobile app, maintaining consistency and reliability. It also supports the goal of real-time monitoring and data-driven decision-making for efficient parking management.

FOL Representation:

$$\forall x (\text{ParkingSpace}(x) \wedge \text{StatusChanged}(x) \rightarrow \text{UpdateDatabase}(x))$$

For every parking space, if a status change is detected, the system updates the database with the new information.

5. Notify Driver When the Parking is Full

IF available_spaces = 0 THEN notify_driver(parking_full)

Explanation:

When all parking spaces in a parking lot are occupied (available spaces equals zero), the AI system automatically sends a notification to drivers approaching or searching for parking in that area. This immediate alert informs users that the parking lot is full, preventing them from wasting time driving to a location with no available spots. The notification allows drivers to quickly search for alternative parking areas nearby, reducing unnecessary traffic and frustration. This knowledge rule supports the goal of reduced search time and enhanced user experience by providing timely information that helps drivers make better decisions.

FOL Representation:

$\forall x (\text{ParkingLot}(x) \wedge \text{AvailableSpaces}(x) = 0 \rightarrow \text{NotifyDriver}(x, \text{ParkingFull}))$

For every parking lot, if the number of available spaces equals zero, then notify driver that the parking is full.

Knowledge Representation Involved to Achieve Goals

The AI Smart Parking Detection System's Knowledge Representations (KRs) work together to accomplish the project's primary objectives, which include effective parking detection, shorter search times, less traffic and improved user experience. Every KR has a distinct function in facilitating data accuracy, real-time updates and intelligent reasoning.

Goal 1: Efficient Parking Detection

Related KRs:

KR1: Detect Occupied Parking Space

KR2: Detect Available Parking Space

Explanation:

Using image recognition, these KRs allow the AI system to precisely determine each parking spot's condition. The area is indicated as "Occupied" when a car is spotted and as "Available" otherwise. The system keeps an accurate and current record of parking space availability thanks to its logical structure.

Goal 2: Reduced Search Time

Related KRs:

KR3: Provide Nearest Available Parking Spot

Explanation:

This KR enables the system to evaluate every place that is available and suggest the one that is closest to the driver's current location. The AI helps drivers save time and minimizes needless driving around the parking lot by automatically displaying the nearest available space.

Goal 3: Real-Time Information and Data Consistency

Related KRs:

KR4: Parking Database Update

Explanation:

This KR makes sure that the database is updated right away whenever a parking space's occupancy status changes. In order to meet the objective of real-time parking detection and enhance overall system reliability, this keeps the detection cameras and the mobile app in sync.

Goal 4: Enhanced User Experience

Related KRs:

KR3: Provide Nearest Available Parking Spot

KR4: Parking Database Update

Explanation:

These KRs guarantee that users get precise, up-to-date information and practical directions to the closest parking space. The smooth integration of navigation, data update and detection improves user pleasure and fosters system confidence.

Goal 5: Decreased Traffic Congestion

Related KRs:

KR5: Notify Driver When the Parking is Full

Explanation:

This KR helps reduce traffic congestion by preventing drivers from entering full parking lots. When no parking spaces are available, the system immediately notifies approaching drivers through the mobile app. This allows drivers to search for alternative parking locations before arriving, avoiding unnecessary circling and reducing traffic buildup at parking entrances. This supports the goal of decreased traffic congestion and improved overall urban traffic flow.

When combined, these Knowledge Representations give the AI system the ability to reason logically, make precise judgments and behave intelligently in order to accomplish its primary goals, which include effective detection, less traffic and a more enjoyable driving experience for all users.

